Question 1: lowk wrap these all in a proof env type splish

(a) We induct over n.

BC We compute that when $n = 1$, $f_{4n} = f_4 = f_3 + f_2 = 2 + 1 = 3$, which is a multiple of 3.

IS Assume that f_{4n} is a multiple of 3. Consider $f_{4(n+1)} = f_{4n+4}$. We compute

$$f_{4n+4} = f_{4n+3} + f_{4n+2} = 2 \cdot f_{4n+2} + f_{4n+1} = 3 \cdot f_{4n+1} + f_{4n}.$$

We can see that we $f_{4(n+1)}$ can be expressed as a sum of a multiple of 3 and f_{4n}, which is a multiple of 3 by assumption. Thus, $f_{4(n+1)}$ is also a multiple of 3.

(b) We induct over n.

BC We compute that when $n = 1$, $f_{5n} = f_5 = f_4 + f_3 = 5$, which is a multiple of 5.

IS Assume f_{5n} is a multiple of 5. Cos

(c) We induct over n.

BC ok

IS Assume $f_1 + f_2 + \cdots + f_{n-1} = \sum_{k=1}^{n-1} f_k = f_{n+1} - 1$. Consider $\sum_{k=1}^{(n-1)+1} f_k$. We then have

$$f_1 + f_2 + \cdots + f_{(n+1)-1} = f_1 + f_2 + \cdots + f_{n-1} + f_n.$$

The sum of the first $n - 1$ terms evaluate to $f_{n+1} - 1$ by assumption. So, we can rewrite as

$$f_{n+1} - 1 + f_n = f_{n+2} - 1.$$

Thus, we are done.

(d) We induct over n.

BC ok

IS Assume $f_1 + f_3 + \cdots + f_{2n-1} = \sum_{k=1}^{n} f_{2k-1} = f_{2n}$. Consider $\sum_{k=1}^{n+1} f_{2k-1}$. Using our assumption, we can rewrite this as

$$\sum_{k=1}^{n+1} f_{2k-1} = \sum_{k=1}^{n} f_{2k-1} + f_{2n+1} = f_{2n} + f_{2n+1}.$$

We see that $f_{2n} + f_{2n+1} = f_{2(n+1)}$. Thus, we are done.

(e) We induct over n.

BC ok

IS Assume that $f_2 + f_4 + \cdots + f_{2n} = \sum_{k=1}^{n} f_{2k} = f_{2n+1} - 1$. Consider $\sum_{k=1}^{n+1} f_{2k}$. Using our assumption, we can rewrite this as

$$\sum_{k=1}^{n+1} f_{2k} = \sum_{k=1}^{n} f_{2k} + f_{2n+2} = f_{2n+1} - 1 + f_{2n+2}.$$

We see that $f_{2n+1} - 1 + f_{2n+2} = f_{2n+3} - 1 = f_{2(n+1)+1} - 1$. Thus, we are done.

(f) We induct over n.

BC ok

IS Assume that ${f_1}^2 + {f_2}^2 + \cdots + {f_n}^2 = \sum_{k=1}^{n} {f_k}^2 = f_n f_{n+1}$. Consider $\sum_{k=1}^{n+1} {f_k}^2$. Using our assumption, we can rewrite this as

$$\sum_{k=1}^{n+1} {f_k}^2 = \sum_{k=1}^{n} {f_k}^2 + {f_{n+1}}^2 = f_n f_{n+1} + {f_{n+1}}^2.$$

We can rewrite this as $f_{n+1}(f_n + f_{n+1}) = f_{n+1} f_{n+2}$. Thus, we are done.

(g) *Proof.* We induct over n.

BC ok

IS Assume $n \not\equiv 0 \pmod 3$, and f_n is odd. If $n \not\equiv 0 \pmod 3$, there are two cases: $n \equiv 1 \pmod 3$ or $n \equiv 2 \pmod 3$. We need not consider both cases; we know that for any n such that $n \not\equiv 0 \pmod 3$, $n + 3 \not\equiv 0 \pmod 3$. WLOG, consider f_{n+3}. We can rewrite this as

$$f_{n+3} = f_{n+2} + f_{n+1} = 2 \cdot f_{n+1} + f_n.$$

By assumption, f_n is odd, and thus f_{n+3} is the sum of an even and an odd, which is always odd.

Thus, we are done. □

Question 2: (a) Using the Euclidean Algorithm, we find that $\gcd(775, 682) = 31 = 8 \cdot 682 - 7 \cdot 775$.

(b) Similarly, we find that $\gcd(562, 243) = 1 = 18 \cdot 562 + 37 \cdot 243$.